

Homework 5

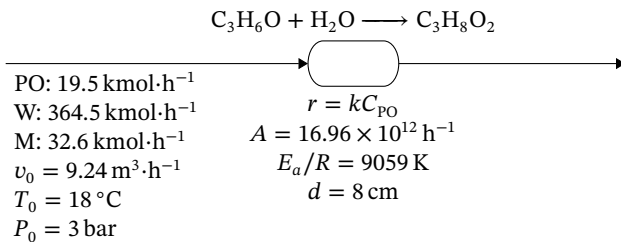
CENG 113, SP 2026

Dr. Drews

Academic Integrity: For all problems **show your work** to receive credit. By submitting your assignment you're confirming that your submission is a product of your own effort.

Unlike previous assignments please **submit the code itself** to Gradescope. Each problem should be a single script file so that when the graders run the script the appropriate plots are produced. Each script should be called `problem1.m`, `problem2.m`, etc. Submissions will be checked with MOSS (Measure of Software Similarity) for originality.

All problems in this assignment concern the **non-elementary**, liquid-phase production of propylene glycol (PG) by hydrolysis of propylene oxide (PO) with water (W) in the presence of methanol (M):



Species	C_p ($\text{J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$)	ΔH_f° ($\text{kJ} \cdot \text{mol}^{-1}$)
propylene oxide	146.4	-122.6
water	75.3	-285.8
propylene glycol	192.4	-501.0
methanol	81.6	-238.7

PO is dissolved in methanol (M, an inert) to lower its viscosity and allow for easier pumping. A small amount of H_2SO_4 is dissolved in the water as a catalyst but its role will be assumed negligible outside producing the rate noted above. The heat transfer coefficient (h) and viscosity (μ) of process stream, if needed, are assumed to have the same properties as liquid water.

Problem 1

Assume the reactor is an adiabatic PFR **with** pressure drop. Determine the reactor volume needed to achieve either 80% conversion of PO or an outlet pressure equal to 2.5 bar, whichever occurs first.

Use a 1-by-3 subplot figure to plot conversion, temperature, and pressure profiles for the length you determined. Label all axes appropriately. (ANS: Outlet $T \approx 60^\circ\text{C}$)

Problem 2

Assume the reactor is a **co-flow**, jacketed PFR **without** pressure drop. The utility is liquid water initially at 35°C and $6.5 \text{ L} \cdot \text{s}^{-1}$. Determine the maximum possible conversion of PO if the utility water is not allowed to exceed 40°C .

Use a 1-by-2 subplot figure to plot the conversion profile on the left, and the two temperature profiles—process **and** utility—on the right. **As a comment at the top of your code** provide a brief description of what the two temperature profiles imply about the role of the utility in this reactor. (ANS: $\approx 70\%$ conversion)

Problem 3

Repeat Problem 2 if the utility is in **counter-flow** orientation. You do not need to add the comment at the top of the code. A suggested approach to this problem is:

- Pick a reactor volume.
- Use the shooting method to determine the appropriate utility temperature.
- Look at the utility temperature profile. If the maximum utility temperature is $40 \pm 0.5^\circ\text{C}$ (i.e., “near” 40°C) then you’re done. Otherwise, pick another reactor volume and repeat.

(ANS: 40-50% conversion but with less volume vs. Problem 2)